Final Project Proposal

STAT 359: Introduction to Large Language Models

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<Abstract>

This project evaluates automated ICD-10 code classification from unstructured clinical notes using pre-trained language models (PLMs). I fine-tune and compare a general BERT model and a domain-specific BioBERT on the same dataset to measure differences in classification performance. In addition, I analyze tokenization differences on medical terminology to help explain performance gaps between the models.

Project Title

Comparative Analysis of Clinical Note Classification: General BERT vs. BioBERT for Automated ICD-10 Coding

Introduction

Converting clinical notes into standardized ICD-10 codes is essential but remains labor-intensive. Although large language models perform well on general text, they often struggle with medical terminology and clinical shorthand. This project benchmarks BioBERT against BERT for ICD-10 coding using the Hugging Face framework, focusing on both classification performance and tokenization differences.

Research Questions & Hypotheses

- **RQ 1 (Performance):** Does domain-specific pre-training improve ICD-10 classification performance compared to a general BERT baseline?
  - **Hypothesis:** BioBERT will achieve a higher Macro F1-score than BERT due to its biomedical pre-training.
- **RQ 2 (Mechanism):** How does pre-training data affect tokenization of medical terminology?
  - **Hypothesis:** BioBERT will produce fewer sub-tokens for medical terms, whereas BERT will fragment them more frequently, which may influence classification performance.

Data & Models (Hugging Face)

- **Dataset:** `birgermoell/icd10-clinical-notes` (or a functionally equivalent clinical dataset)
  This dataset contains physician-written clinical notes paired with ICD-10 codes.
- **Baseline Model:** `bert-base-uncased` (or similar)

A general-domain PLM trained on Wikipedia and BookCorpus.

- **Target Model:** `dmis-lab/biobert-v1.1-base` (or similar)
  A model further pre-trained on biomedical corpora such as PubMed.

Methodology & Analysis Plan

- **Exploratory Data Analysis (EDA):** Analyze the distribution of ICD-10 codes to identify class imbalance and understand dataset characteristics.
- **Tokenization Study:** Compare how BERT and BioBERT tokenize representative medical terms (e.g., "Appendicitis," "Myocardial Infarction") by examining the number of sub-tokens produced.
- **Fine-tuning:** Fine-tune both models using the Hugging Face Trainer API under identical hyperparameters (learning rate, batch size, and number of epochs) to ensure a fair comparison.
- **Evaluation & Error Analysis:** Evaluate performance using Accuracy and Macro/Weighted F1-scores to account for class imbalance. Use a confusion matrix to identify common misclassified disease categories and compare model error patterns.

Expected Results & Implications

I expect BioBERT to outperform BERT in Macro/Weighted F1 on ICD-10 classification. I will use tokenization and confusion-matrix analyses to help explain the performance gap. This will inform model selection for biomedical text classification tasks.

Hardware & Environment

- CPU: 13th Gen Intel(R) Core (TM) i7-13700H
- RAM: 32.0GB
- GPU: NVIDIA GeForce RTX 4050 Laptop GPU